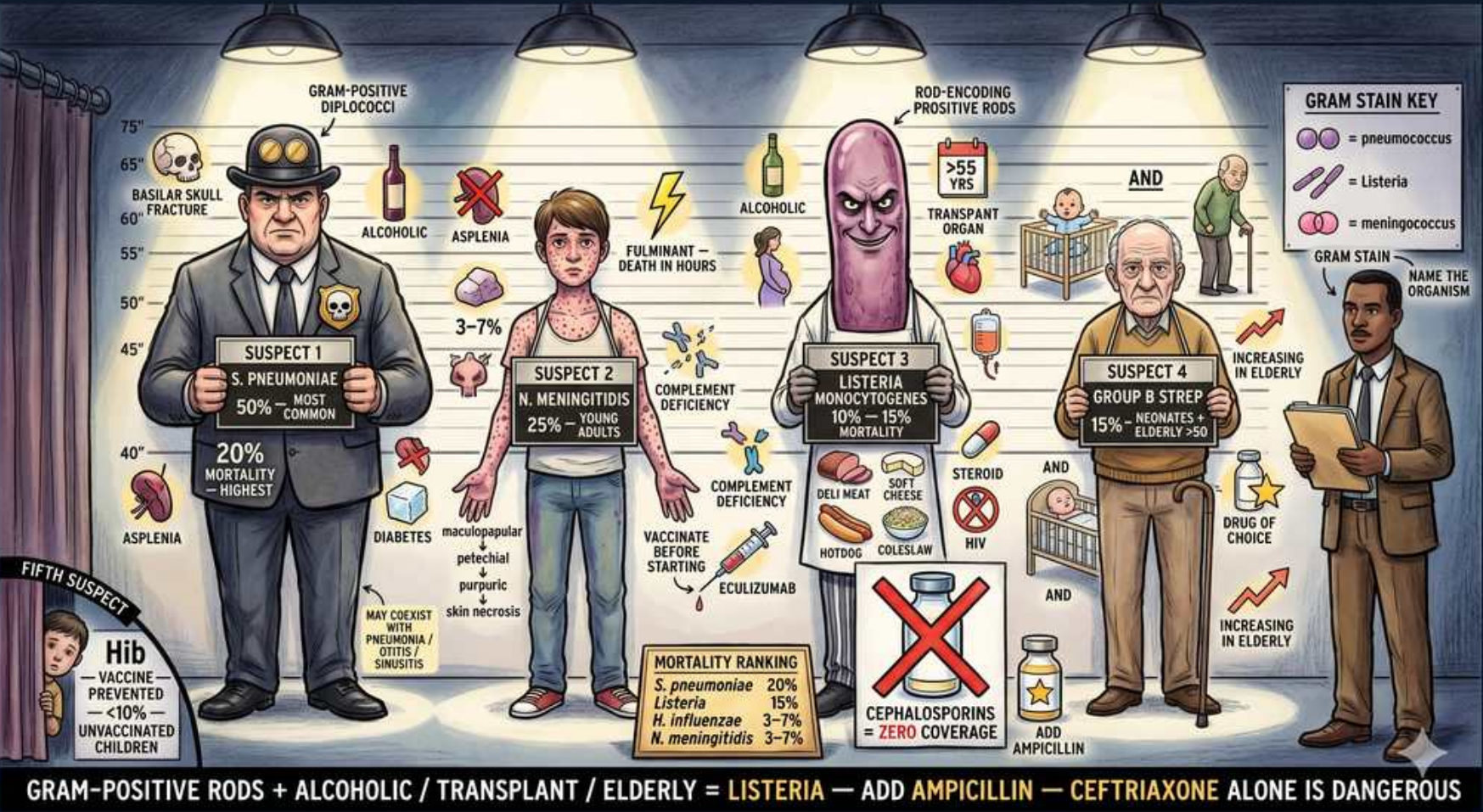


Bacterial Meningitis: Pathogen Lineup by Host & Stain



1. S. pneumoniae (Suspect 1)

WHAT YOU SEE

Detective figure, gram-positive diplococci, 50% sign

WHAT IT ENCODES

Most common cause of adult community bacterial meningitis (~50%) and the deadliest (~20% mortality). Gram-positive diplococci. Risk rises with asplenia, basilar skull fracture/CSF leak, and alcoholism. Covered by vancomycin + ceftriaxone.

BOARD TRAP

Pneumococcus, not meningococcus, is the leading adult cause and carries the highest mortality — don't pick N. meningitidis just because of the 'meningitis' name.

2. Basilar skull fracture/CSF leak

WHAT YOU SEE

Skull icon above Suspect 1

WHAT IT ENCODES

A basilar skull fracture or persistent CSF rhinorrhea/otorrhea predisposes to recurrent pneumococcal meningitis via direct nasopharyngeal-CSF communication.

BOARD TRAP

Recurrent bacterial meningitis with a CSF leak history = S. pneumoniae — investigate for an anatomic defect, not immunodeficiency first.

3. Asplenia

WHAT YOU SEE

Spleen icon, Suspect 1

WHAT IT ENCODES

Asplenia removes clearance of encapsulated organisms, raising risk of pneumococcal (and meningococcal/Hib) sepsis and meningitis. These patients need encapsulated-organism vaccines.

BOARD TRAP

Asplenia predisposes to ALL encapsulated organisms — but the most common resulting meningitis is still pneumococcal.

4. N. meningitidis (Suspect 2)

WHAT YOU SEE

Young adult figure, 25% sign, petechial rash

WHAT IT ENCODES

Causes ~25% of meningitis in young adults; spreads in dorms/military. Gram-negative diplococcus with petechial/purpuric rash and Waterhouse-Friderichsen adrenal hemorrhage. Terminal complement (C5-C9) deficiency predisposes.

BOARD TRAP

A purpuric/petechial rash with meningitis points to meningococcus — recurrent Neisseria infections specifically signal terminal complement deficiency, not generic immunodeficiency.

5. Complement deficiency

WHAT YOU SEE

Suspect 2, complement label, eculizumab

WHAT IT ENCODES

Terminal complement (C5-9) or properdin deficiency causes recurrent Neisseria infections. Patients on eculizumab share this risk and must receive meningococcal vaccination before therapy.

BOARD TRAP

Recurrent Neisseria = terminal complement defect — and eculizumab (anti-C5) mimics it, mandating meningococcal vaccine prophylaxis.

6. L. monocytogenes (Suspect 3)

WHAT YOU SEE

Bald villain, gram-positive rod, 10-15% mortality

WHAT IT ENCODES

Gram-positive rod causing meningitis in neonates, pregnant women, the elderly (>50), and immunocompromised/steroid/transplant patients. From deli meat, soft cheese, hot dogs. 10-15% mortality.

BOARD TRAP

Listeria is the reason to ADD ampicillin in patients >50, immunocompromised, or pregnant — cephalosporins have ZERO Listeria coverage.

7. Listeria food sources

WHAT YOU SEE

Deli meat / soft cheese / hot dog icons

WHAT IT ENCODES

Listeria is acquired from unpasteurized soft cheese, deli meats, hot dogs, and refrigerated ready-to-eat foods — it grows at refrigeration temperature (cold enrichment).

BOARD TRAP

A pregnant patient with febrile illness after deli meat/soft cheese should be treated empirically for Listeria with ampicillin.

8. Cephalosporins = zero Listeria

WHAT YOU SEE

Crossed-out cephalosporin, ADD ampicillin

WHAT IT ENCODES

All cephalosporins (including ceftriaxone) have no activity against Listeria. Ampicillin (or TMP-SMX if penicillin-allergic) must be added for at-risk hosts.

BOARD TRAP

Ceftriaxone alone is inadequate in the elderly/immunocompromised — forgetting to add ampicillin for Listeria is the classic board miss.

9. Group B Strep (Suspect 4)

WHAT YOU SEE

Elderly figure, 15% in elderly >50

WHAT IT ENCODES

Streptococcus agalactiae (GBS) causes neonatal meningitis (early/late onset) and is increasingly seen in elderly/diabetic adults (~15%). Gram-positive cocci in chains.

BOARD TRAP

GBS is rising in the elderly and diabetics, not just neonates — don't restrict it to newborns on a geriatric meningitis question.

10. Increasing in elderly

WHAT YOU SEE

Arrow up, elderly tag Suspect 4

WHAT IT ENCODES

Both Listeria and Group B Strep incidence climb with age >50, which is exactly why ampicillin is added to vancomycin + ceftriaxone in older adults.

BOARD TRAP

Age >50 expands empiric coverage to include Listeria/GBS — the regimen is not the same as for a healthy young adult.

11. Hib (5th suspect)

WHAT YOU SEE

Bottom-left small figure, vaccine-preventable

WHAT IT ENCODES

Haemophilus influenzae type b now causes <10% of meningitis, mainly in unvaccinated children. Gram-negative coccobacillus. Conjugate vaccine dramatically reduced incidence.

BOARD TRAP

Hib meningitis is now a disease of UNVACCINATED children — its rarity in the vaccinated era is itself a board point.

12. Mortality ranking

WHAT YOU SEE

Suspect 3 placard, mortality table

WHAT IT ENCODES

Relative meningitis mortality: S. pneumoniae ~20% (highest), Listeria ~15%, H. influenzae ~7%, N. meningitidis ~3-7%. Pneumococcus is both most common and most lethal in adults.

BOARD TRAP

Meningococcus has dramatic presentation but LOWER mortality than pneumococcus — fulminant appearance does not equal highest death rate.

13. Fulminant meningococcemia

WHAT YOU SEE

Lightning bolt, death-in-hours label

WHAT IT ENCODES

Meningococcemia can kill within hours via DIC and Waterhouse-Friderichsen syndrome (bilateral adrenal hemorrhage). Demands immediate antibiotics and droplet precautions plus close-contact prophylaxis.

BOARD TRAP

Don't delay antibiotics for LP/imaging in suspected meningococcemia — empiric treatment and prophylaxis of contacts cannot wait.

14. Empiric regimen banner

WHAT YOU SEE

Bottom banner, ampicillin + ceftriaxone

WHAT IT ENCODES

For gram-positive rods / alcoholic / transplant / elderly hosts, add ampicillin for Listeria; ceftriaxone alone is dangerous because it misses Listeria. Add vancomycin for resistant pneumococcus.

BOARD TRAP

The banner's whole point: ceftriaxone alone is insufficient in Listeria-risk hosts — the triad is vancomycin + ceftriaxone + ampicillin.